6.9 Package thermal characteristics

The maximum chip-junction temperature, T_J max, in degrees Celsius, can be calculated using the following equation:

$$T_J \max = T_A \max + \left(P_D \max \times \Theta_{JA}\right)$$

Where:

- T_A max is the maximum ambient temperature in °C.
- Θ_{JA} is the package junction-to-ambient thermal resistance in °C/W.
- P_D max is the sum of P_{INT} max and $P_{I/O}$ max:

$$P_D \max = P_{INT} \max + P_{I/O} \max)$$

- P_{INT} max is the product of I_{DD} and V_{DD}, expressed in Watts. This is the maximum chip internal power.

$P_{I/O}$ max represents the maximum power dissipation on output pins:

$$P_{I/O} \max = \sum\left(V_{OL} \times I_{OL}\right) + \sum\left(\left(V_{DDIOx} - V_{OH}\right)\right) \times I_{OH}$$

taking into account the actual V_{OL}/I_{OL} and V_{OH}/I_{OH} of the I/Os at low and high level in the application.

Table 84. **Package thermal characteristics**

Symbol	Definition	Parameter	Value			Unit
			junction-ambient Θ_{JA}	junction-board Θ_{JB}	junction-case Θ_{JC}	
Θ	Thermal resistance	LQFP144	39.0	27.8	10.3	°C/W
		LQFP100	38.0	24.0	10.4	
		LQFP80	41.0	25.3	11.7	
		LQFP64	42.6	24.9	12.1	
		LQFP48	49.6	26.9	14.6	
		LQFP32	49.6	26.9	14.6	
		UFQFPN48	30.1	14.4	9.7	
		UFQFPN32	38.2	19.9	19.5	

6.9.1 Reference documents

- JESD51-2 Integrated Circuits Thermal Test Method Environment Conditions - Natural Convection (Still Air) available on www.jedec.org.
- For information on thermal management, refer to application note *"Guidelines for thermal management on STM32 applications" (AN5036)* available on www.st.com.